



GATE ME 90 Days Challenge

Day 1/90 · Practice Set with Detailed Solutions

SUBJECT

Engineering Mechanics

TODAY'S TOPIC

Statics, Trusses & Friction

COUNTDOWN

110 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on free-body diagrams, equilibrium, trusses and frames, and friction. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 8 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

Three coplanar forces act at a point: 50 N along the +x axis, 80 N at 60° and 60 N at 150° (angles measured anticlockwise from +x). The magnitude of their resultant is

(A) 106.3 N

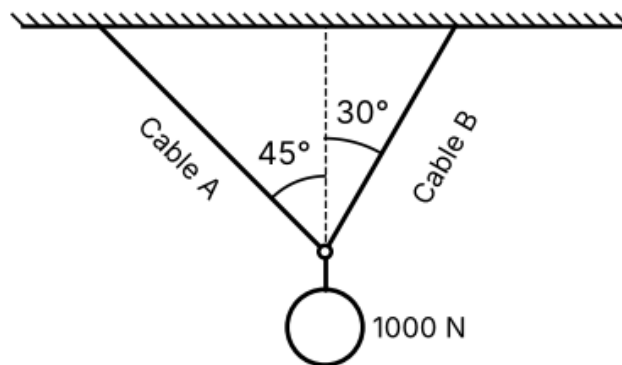
(B) 190.0 N

(C) 99.3 N

(D) 121.4 N

Q2 (NAT)

A 1000 N lamp is hung from two cables as shown. Cable A makes 45° and cable B makes 30° with the vertical. The tension in cable B (in N) is _____



Q3 (MCQ)

A 200 N force is applied at the end of a 0.5 m spanner, making 60° with the handle. The moment about the bolt centre is

(A) 100.0 N·m

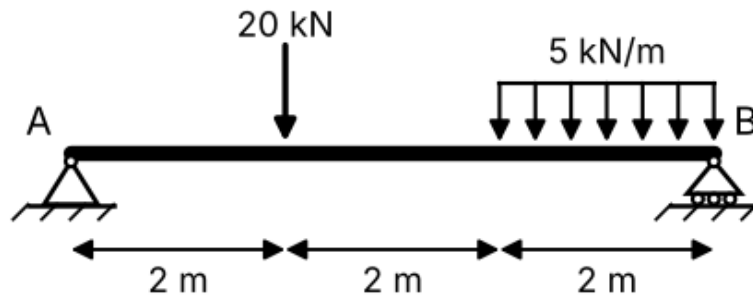
(B) 86.6 N·m

(C) 50.0 N·m

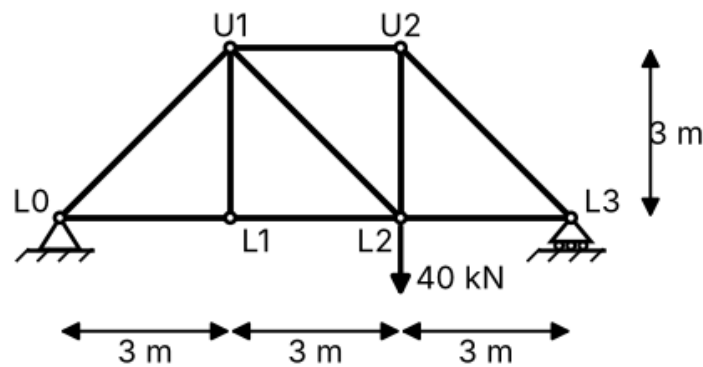
(D) 173.2 N·m

Q4 (NAT)

The simply supported beam AB carries a 20 kN point load and a 5 kN/m UDL as shown. The reaction at support B (in kN) is _____

**Q5** (MCQ)

The truss shown carries a single load of 40 kN at joint L2. L0 is a pin and L3 is a roller. Which member is a zero-force member?



(A) U1L1

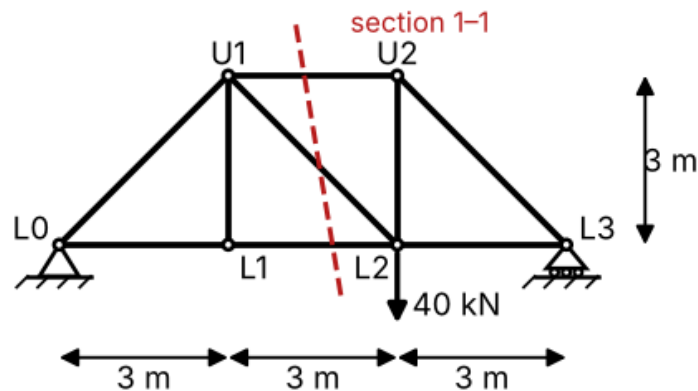
(B) U2L2

(C) U1L2

(D) L1L2

Q6 (NAT)

For the same truss (40 kN at L2), the force in the top chord member U1U2 (magnitude, in kN) is _____. Also state whether it is tension or compression.



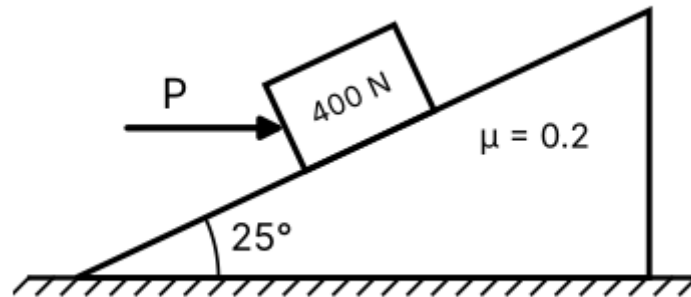
Q7 (MCQ)

A 500 N block on a rough horizontal floor ($\mu = 0.3$) is pulled by a force P inclined at θ above the horizontal. The least value of P that can start the motion, and the angle at which it acts, are

- (A) 143.7 N at 16.7° (B) 150.0 N at 0°
 (C) 143.7 N at 73.3° (D) 137.6 N at 16.7°

Q8 (NAT)

A 400 N block rests on a rough plane inclined at 25° ($\mu = 0.2$). The horizontal force P needed to just push it up the plane (in N) is _____

**Q9** (MCQ)

A uniform ladder rests against a rough vertical wall and on a rough floor, with $\mu = 0.25$ at both surfaces. The least angle the ladder can make with the floor without slipping is

- (A) 61.9° (B) 63.4°
 (C) 75.9° (D) 56.3°

Q10 (NAT)

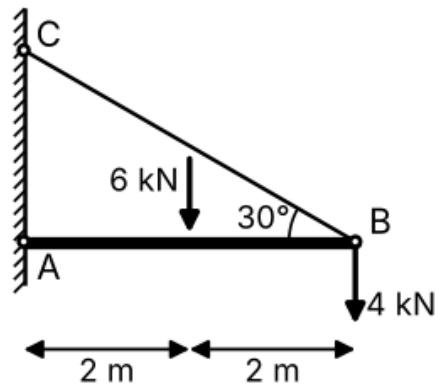
A rope passes half-way round a fixed rough cylinder ($\mu = 0.3$) and is used to raise a load of 500 N slowly. The pull needed on the other end (in N) is _____

Q11 (NAT)

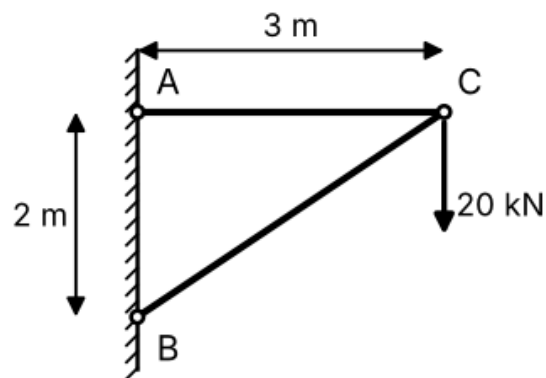
Blocks A (200 N, $\mu = 0.2$) and B (300 N, $\mu = 0.3$) are tied together by a string parallel to an inclined plane, with A below B. The angle of inclination (in degrees) at which the pair just begins to slide down is _____

Q12 MCQ

A light horizontal beam AB, hinged to the wall at A, is held by a cable BC at 30° to the beam. It carries 6 kN at mid-span and 4 kN at B. The tension in the cable is

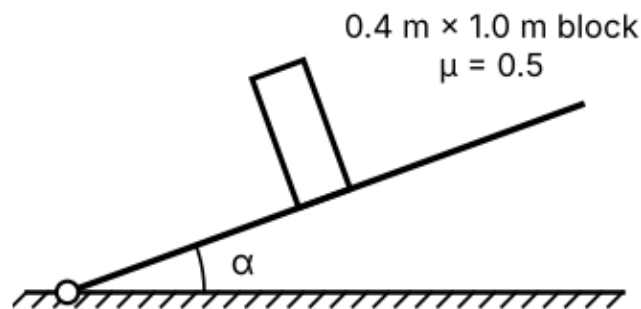
**(A)** 14.0 kN**(B)** 7.0 kN**(C)** 28.0 kN**(D)** 16.2 kN**Q13** MCQ

For the wall-mounted truss shown, with 20 kN hanging at C, the force in member AC is

**(A)** 30 kN (tension)**(B)** 30 kN (compression)**(C)** 36.06 kN (compression)**(D)** 20 kN (tension)

Q14 (MCQ)

A uniform block 0.4 m wide and 1.0 m tall stands on a plane whose inclination α is slowly increased. If $\mu = 0.5$, the block will



(A) Tip over at $\alpha = 21.8^\circ$

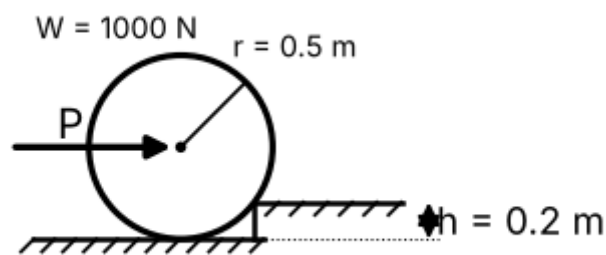
(B) Slide at $\alpha = 26.6^\circ$

(C) Tip over at $\alpha = 26.6^\circ$

(D) Slide at $\alpha = 21.8^\circ$

Q15 (NAT)

A roller of weight 1000 N and radius 0.5 m is to be pulled over a step 0.2 m high by a horizontal force P through its centre. The least P needed (in N) is _____



Answer Key

Q	Type	Answer
Q1	MCQ	(A) 106.3 N, at 69.0° to the +x axis
Q2	NAT	732 (accept 730 to 734)
Q3	MCQ	(B) 86.6 N·m
Q4	NAT	15
Q5	MCQ	(A) U1L1
Q6	NAT	26.67 compression (accept 26.5 to 26.8)
Q7	MCQ	(A) 143.7 N at 16.7°
Q8	NAT	294 (accept 292 to 296)
Q9	MCQ	(A) 61.9°
Q10	NAT	1283 (accept 1275 to 1290)
Q11	NAT	14.57 (accept 14.4 to 14.8)
Q12	MCQ	(A) 14.0 kN
Q13	MCQ	(A) 30 kN (tension)
Q14	MCQ	(A) Tip over at $\alpha = 21.8^\circ$
Q15	NAT	1333 (accept 1320 to 1345)

Detailed Solutions

Q1 Answer: (A) 106.3 N, at 69.0° to the +x axis

Concept: Resolve every force into x and y components, add them, then combine.

$$\Sigma F_x = 50 + 80 \cos 60^\circ + 60 \cos 150^\circ = 50 + 40 - 51.96 = \mathbf{38.04 \text{ N}}$$

$$\Sigma F_y = 0 + 80 \sin 60^\circ + 60 \sin 150^\circ = 69.28 + 30 = \mathbf{99.28 \text{ N}}$$

$$R = \sqrt{(38.04^2 + 99.28^2)} = \mathbf{106.3 \text{ N}}; \text{ direction } \theta = \tan^{-1}(99.28/38.04) = \mathbf{69.0^\circ} \text{ from } +x.$$

Exam tip: 190 N is the arithmetic sum, a common trap. Forces add as vectors, never as numbers.

Q2 Answer: 732 (accept 730 to 734)

Concept: Three concurrent forces in equilibrium. Write $\Sigma F_x = 0$ and $\Sigma F_y = 0$ (or use Lami's theorem).

$$\text{Horizontal: } T_A \sin 45^\circ = T_B \sin 30^\circ \rightarrow T_A = 0.7071 T_B$$

$$\text{Vertical: } T_A \cos 45^\circ + T_B \cos 30^\circ = 1000 \rightarrow 0.5 T_B + 0.866 T_B = 1000$$

$$T_B = 1000 / 1.366 = \mathbf{732.1 \text{ N}}, \text{ and } T_A = \mathbf{517.6 \text{ N}}.$$

$$\mathbf{Lami check: } T_B / \sin(180^\circ - 45^\circ) = 1000 / \sin(75^\circ) \rightarrow T_B = 1000 \times 0.7071 / 0.9659 = 732 \text{ N } \checkmark$$

Exam tip: The cable closer to the vertical always carries more load.

Q3 Answer: (B) 86.6 N·m

Concept: Moment = force $\times$ perpendicular distance, or use only the force component perpendicular to the arm (Varignon's theorem).

Perpendicular component = $200 \sin 60^\circ = 173.2 \text{ N}$

$$M = 173.2 \times 0.5 = \mathbf{86.6 \text{ N}\cdot\text{m}}$$

Exam tip: 100 N·m would be correct only if the force were at 90° to the handle.

Q4 Answer: 15

Concept: Replace the UDL by its resultant at the centre of its length, then take moments about A.

UDL resultant = $5 \times 2 = 10 \text{ kN}$, acting at 5 m from A.

$$\Sigma M_A = 0: R_B \times 6 = 20 \times 2 + 10 \times 5 = 90 \rightarrow R_B = \mathbf{15 \text{ kN}}$$

$$\Sigma F_y = 0: R_A = 30 - 15 = \mathbf{15 \text{ kN}}$$

Exam tip: Always check the second reaction with $\Sigma F_y = 0$; it catches arithmetic slips.

Q5 Answer: (A) U1L1

Concept: If three members meet at an unloaded joint and two of them are collinear, the third member carries zero force.

At joint L1, members L0L1 and L1L2 are collinear (both horizontal) and no load acts at L1.

So $\Sigma F_y = 0$ at L1 gives $F_{U1L1} = \mathbf{0}$.

U2L2 is not zero, because the 40 kN load acts at L2.

Exam tip: Spot zero-force members first. They shrink the problem before you write any equation.

Q6 Answer: 26.67 compression (accept 26.5 to 26.8)

Concept: Method of sections. Cut through U1U2, U1L2 and L1L2 (section 1-1) and take moments about the point where two of the cut members meet.

Reactions: $R_{L0} = 40 \times 3/9 = \mathbf{13.33 \text{ kN}}$, $R_{L3} = \mathbf{26.67 \text{ kN}}$.

Left part, moments about L2 (where U1L2 and L1L2 meet): $R_{L0} \times 6 + F_{U1U2} \times 3 = 0$, taking F as tension.

$$F_{U1U2} = -13.33 \times 6 / 3 = \mathbf{-26.67 \text{ kN} \rightarrow 26.67 \text{ kN compression.}}$$

Exam tip: The top chord of a simply supported truss under downward loads is always in compression; the bottom chord is in tension.

Q7 Answer: (A) 143.7 N at 16.7°

Concept: $P \cos \theta = \mu(W - P \sin \theta) \rightarrow P = \mu W / (\cos \theta + \mu \sin \theta)$. P is least when the denominator is greatest, i.e. when $\tan \theta = \mu$ (θ equals the angle of friction).

$$\theta = \tan^{-1}(0.3) = \mathbf{16.7^\circ}$$

$$P_{\min} = \mu W / \sqrt{1 + \mu^2} = 150 / 1.044 = \mathbf{143.7 \text{ N}}$$

Exam tip: Pulling slightly upward beats a horizontal pull (150 N) because it reduces the normal reaction.

Q8 Answer: 294 (accept 292 to 296)

Concept: With the angle of friction $\phi = \tan^{-1}\mu$, a horizontal force to push a block up an incline is $P = W \tan(\alpha + \phi)$.

$$\phi = \tan^{-1}(0.2) = 11.31^\circ, \text{ so } \alpha + \phi = 36.31^\circ$$

$$P = 400 \times \tan 36.31^\circ = 400 \times 0.7348 = \mathbf{293.9 \text{ N}}$$

Check by components: along the plane $P \cos \alpha = W \sin \alpha + \mu N$, with $N = W \cos \alpha + P \sin \alpha$. Solving gives the same 293.9 N.

Exam tip: A horizontal push also presses the block into the plane, which increases friction. That is why it needs more force than a push along the plane.

Q9 Answer: (A) 61.9°

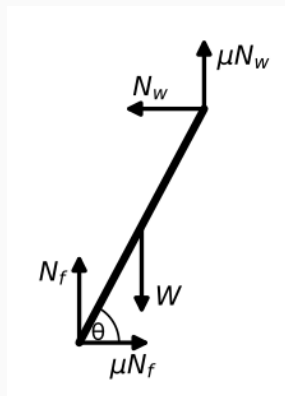
Concept: At impending slip both friction forces reach their limits: $F_f = \mu N_f$ (towards the wall) and $F_w = \mu N_w$ (upwards).

$$\Sigma F_x: N_w = \mu N_f, \Sigma F_y: N_f + \mu N_w = W \rightarrow N_f = W/(1 + \mu^2)$$

$$\Sigma M \text{ about the foot: } N_w L \sin \theta + \mu N_w L \cos \theta = W (L/2) \cos \theta$$

$$\text{This simplifies to } \tan \theta = (1 - \mu^2)/(2\mu) = (1 - 0.0625)/0.5 = 1.875 \rightarrow \theta = \mathbf{61.9^\circ}$$

Exam tip: With a smooth wall the formula becomes $\tan \theta = 1/(2\mu) = 2$, giving 63.4° . Wall friction lets the ladder stand slightly flatter.

**Q10 Answer: 1283 (accept 1275 to 1290)**

Concept: Belt (capstan) friction: $T_1/T_2 = e^{\mu\theta}$, with T_1 the tight side and θ in radians.

$$\theta = \pi \text{ rad (half turn), so } e^{0.3\pi} = e^{0.9425} = 2.566$$

$$\text{To raise the load, the pull is on the tight side: } T_1 = 500 \times 2.566 = \mathbf{1283 \text{ N}}$$

Exam tip: To lower the load slowly, the same cylinder needs only $500/2.566 = 195 \text{ N}$. Always decide which side is tight first.

Q11 Answer: 14.57 (accept 14.4 to 14.8)

Concept: When connected bodies slide together, total driving force = total limiting friction.

$$\text{Driving force} = (200 + 300) \sin \alpha = 500 \sin \alpha$$

$$\text{Total friction} = (0.2 \times 200 + 0.3 \times 300) \cos \alpha = 130 \cos \alpha$$

$$500 \sin \alpha = 130 \cos \alpha \rightarrow \tan \alpha = 0.26 \rightarrow \alpha = \mathbf{14.57^\circ}$$

Exam tip: Neither block alone sets the angle (A alone would slide at 11.3° , B alone at 16.7°). The string makes them share friction.

Q12 Answer: (A) 14.0 kN

Concept: Take moments about the hinge A so that both unknown hinge reactions drop out.

Vertical component of T at B = $T \sin 30^\circ$, acting at 4 m from A.

$$\Sigma M_A = 0: T \sin 30^\circ \times 4 = 6 \times 2 + 4 \times 4 = 28 \rightarrow T = \mathbf{14 \text{ kN}}$$

$$\text{Hinge reaction: } A_x = T \cos 30^\circ = 12.12 \text{ kN}, A_y = 10 - 7 = 3 \text{ kN} \rightarrow R_A = \mathbf{12.49 \text{ kN}}$$

Exam tip: The cable is a two-force member, so its tension acts along the cable. Use that direction directly.

Q13 Answer: (A) 30 kN (tension)

Concept: Method of joints at C, the joint with only two unknown members.

BC has slope 2/3, so $\sin \alpha = 2/\sqrt{13}$ and $\cos \alpha = 3/\sqrt{13}$.

$$\Sigma F_y \text{ at C: } F_{BC} \sin \alpha = 20 \rightarrow F_{BC} = 36.06 \text{ kN, pushing C upward (compression).}$$

$$\Sigma F_x \text{ at C: } F_{AC} = F_{BC} \cos \alpha = 36.06 \times 3/\sqrt{13} = \mathbf{30 \text{ kN tension}}$$

Exam tip: In a wall bracket the upper member is in tension (it is pulled) and the lower strut is in compression (it is pushed).

Q14 Answer: (A) Tip over at $\alpha = 21.8^\circ$

Concept: Compare two limits. Sliding starts when $\tan \alpha = \mu$. Tipping starts when the weight's line of action passes the lower edge, i.e. $\tan \alpha = b/h$.

$$\text{Sliding: } \tan \alpha = 0.5 \rightarrow \alpha = 26.6^\circ$$

$$\text{Tipping: } \tan \alpha = 0.4/1.0 \rightarrow \alpha = 21.8^\circ$$

The smaller angle is reached first, so the block **tips over at 21.8°** before it can slide.

Exam tip: Tall, narrow bodies tip; short, wide bodies slide. Always check both.

Q15 Answer: 1333 (accept 1320 to 1345)

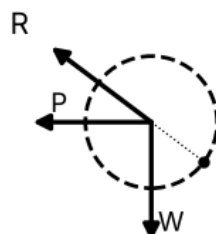
Concept: At the moment of lifting, the floor reaction becomes zero. The roller then has three forces (P, W and the step reaction R), and R passes through the centre.

Take moments about the step corner. Lever arm of W = $\sqrt{(r^2 - (r - h)^2)} = \sqrt{(0.25 - 0.09)} = 0.4 \text{ m}$.

Lever arm of P = $r - h = 0.3 \text{ m}$.

$$P \times 0.3 = 1000 \times 0.4 \rightarrow P = \mathbf{1333 \text{ N}}$$

Exam tip: $P = W\sqrt{(h(2r - h))}/(r - h)$. As h approaches r the force becomes infinite, which is why a horizontal pull cannot lift a roller over a step as high as its radius.



R acts along the line from the step corner to the centre

What next?

Day 1 test tonight, 7 to 8 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: comment GATE90 on our reel for the link.



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Next, Day 2/90: Engineering Mechanics, Kinematics & Dynamics. Comment **GATE90** on our reel to get all links.

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